

Network-Constrained Flexibility Quantification of Heat Pumps with Integrated Space Heating and Storage Tanks in Active Distribution Grids

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Abstract: The rapid growth in Heat Pump (HP) usage poses considerable challenges for distribution grids, yet also unlocks opportunities for more efficient, optimized operation. This paper proposes a secure and tractable method for quantifying and exploiting HP flexibility. The framework adopts a two-stage operational scheduling and flexibility quantification approach that integrates building thermal inertia and Domestic Hot Water (DHW) storage into a convex Optimal Power Flow (OPF) formulation. First, a baseline day-ahead scheduling problem ensures cost-effective HP operation while respecting user comfort and tank temperature limits. Second, upward and downward flexibility envelopes are derived by treating the combined building-tank system as a virtual battery and incorporating both thermal dynamics and network constraints via OPF. The methodology is validated on a realistic low-voltage distribution network with residential loads, photovoltaic systems, and stochastic DHW demand. Results demonstrate that upward flexibility is constrained by comfort and voltage constraints at higher HP penetration levels, whereas downward flexibility is constrained solely by user comfort. Furthermore, the case study highlights that network limits become binding at and beyond 60% HP penetration, underscoring the importance of the approach in realistic high-adoption scenarios.

Keywords: Power Systems and Power Electronics; Electrical Distribution Systems; Heat Pump Flexibility; Space Heating; Domestic Hot Water; Optimal Power Flow

Indices

NOMENCLATURE

k, m	Index of buses.	ϕ_d, ϕ_{hp}	Power factor angle of d and hp .
d, hp, der	Index of basic loads, HPs, and DERs	g_k, b_k	Conductance and susceptance respectively
$\uparrow / \downarrow$	Index of upward / downward flexibility.	r_{km}, x_{km}	Line resistance and reactance
tra, t	Index of transformer and time period.	t^{dep}, t^{arr}	Departure and arrival time.
$m : k \sim m$	Index of line directed from bus k to $m, \forall m$	$\bullet / \underline{\bullet} / \hat{\bullet}$	Upper/Lower/estimated value of a quantity.
$ref, flex$	Index of baseline and flexibility operation case	$v_{s,k,t}$	Squared voltage magnitude of bus k .
		$\ell_{s,km,t}$	Squared current flow from bus k to bus m .

Sets

Ω^B, Ω^L	Set of buses and lines
Ω^d	Set of loads where Ω_k^d indicates set of loads connected to bus k .
Ω^{hp}	Set of HPs where Ω_k^{hp} indicates set of HPs connected to bus k .
Ω^{der}	Set of DERs where Ω_k^{der} indicates set of DERs connected to bus k .
Ω^H	Set of time intervals.
Ω^S	Set of cases: $s = sh$ for space heating flexibility activation, $s = dhw$ for DHW flexibility activation.

Parameters and Variables

P, Q, S	Active, reactive, apparent electrical power.
Q, h	Thermal power and time interval
$\phi_{der}(P)$	Function of power factor angle of der in terms of active power P .

1. INTRODUCTION

The transition from conventional fossil-fuel-based heating to electricity-driven systems, most notably Heat Pumps (HPs), is becoming increasingly inevitable. This shift changes energy consumption patterns, presenting both challenges and opportunities for power system management. Consequently, flexibility and operational scheduling in power systems require innovative approaches to address temporal and spatial power system congestion.

The first limitation concerns the ambiguity in defining flexibility. The concept of HP flexibility is interpreted differently across research contexts and target audiences, and this inconsistency becomes even more evident in quantification methodologies, as discussed by Reynders et al.

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(2018). A second limitation arises from the gap between disciplines: building engineers are often unfamiliar with the technical details of energy networks, and conversely, power system engineers may lack expertise in building dynamics. Therefore, adopting simple, intuitive methodologies for quantifying flexibility for both communities can greatly enhance mutual understanding and collaboration.

Heat pump flexibility encompasses the capability to adjust its operation across different setpoints, enabling both energy optimization and grid support. This flexibility can be exploited to deliver services, including frequency stabilization and congestion alleviation. The inherent thermal inertia of buildings allows for short-term demand shifting through pre-heating or pre-cooling strategies, helping smooth the overall load profile Harb et al. (2016). Furthermore, integrating thermal storage devices, such as hot water tanks or phase-change material reservoirs, supports extended demand response by decoupling the timing of energy consumption from immediate thermal comfort needs, as supported by Baeten et al. (2017); Hirmiz et al. (2019).

Heat pump operation scheduling has been widely exploited in recent literature to mitigate various issues that existing power grids encounter. Following Langner et al. (2024), single buildings result in grid-adverse impact; therefore, multi-building coordination is essential. Consequently, the literature proposes central, rule-based, and hierarchical approaches. A rule-based algorithm that coordinates clusters of buildings for Space Heating (SH) and Domestic Hot Water (DHW) heating is proposed by Langner et al. (2024). A hierarchical control approach is proposed in Bhattarai et al. (2014), where a central controller initially schedules HP operation to minimize overall energy cost, while a local adaptive controller ensures consumer comfort and network voltage requirements. Reynders et al. (2018) quantifies HP flexibility solely through building thermal mass, using the tank solely for DHW, which likely understates the true storage and load-shifting potential of SH.

Following the above literature, two main issues can be identified. First, computational demand increases significantly when multiple building-level algorithms are executed simultaneously. Second, most existing methodologies investigate either the flexibility potential of HPs or their impact on network operations, but rarely consider both simultaneously. This gap is crucial, as flexibility assessment requires jointly considering user comfort and network constraints to enable effective coordination.

Therefore, the contribution of this paper is twofold. First, it highlights the flexibility potential of HPs in active distribution networks. Second, it proposes a novel day-ahead optimization framework for cost-effective HP operation and for quantifying flexibility under user and network constraints.

The remainder of this paper is organized as follows. Section 2 presents the thermal and electrical modeling. Section 3 discusses the flexibility margins of HPs. Section 4 outlines the proposed secured flexibility quantification framework. The case study is described in Section 5. Section 6 reports and discusses the simulation results, and finally, Section 7 provides the conclusions.

2. THERMAL AND ELECTRICAL MODELING

2.1 Building Thermodynamics

Each building is modeled with a reduced-order RC network that combines physics-based structure with data-driven parameter identification, balancing accuracy and computational efficiency as shown in Déqué et al. (2000). A $3R2C$ model is employed, although other RC topologies could be substituted. The state-space equations (1)–(2) describe the temperature dynamics; $R_{i,j}$ [K/W] denotes the thermal resistance between nodes i and j , C_i [J/K] the thermal capacitance of node i , T_a the ambient temperature, Q_{sol} [W/m²] the incident solar irradiance, and Q_{sh}^{hp} [W] the thermal power supplied by the heating/cooling system. The coefficient $f_h \in [0, 1]$ partitions the delivered heat between the indoor air and the envelope mass, while A_{in} and A_e [m²] denote effective solar apertures applied to Q_{sol} for the indoor air and the envelope, respectively.

$$C_{in} \frac{dT_{in}}{dt} = \frac{T_e - T_{in}}{R_{in,e}} + \frac{T_a - T_{in}}{R_{in,a}} + f_h Q_{sh}^{hp} + A_{in} Q_{sol} \quad (1)$$

$$C_e \frac{dT_e}{dt} = \frac{T_{in} - T_e}{R_{in,e}} + \frac{T_a - T_e}{R_{e,a}} + (1 - f_h) Q_{sh}^{hp} + A_e Q_{sol} \quad (2)$$

Calibration relies on four measured or simulated signals over a defined horizon: Q_{sol} , T_a , Q_{sh}^{hp} , T_{in} . Parameters $\{R_{i,j}, C_i, f_h, A\}$ are identified via least squares on the discretized-time model; thereafter, given any three of $\{Q_{sol}, T_a, Q_{sh}^{hp}, T_{in}\}$, the fourth can be inferred during operation. Additional modeling and tuning details are provided by Panagi et al. (2026a).

2.2 Thermal Storage Tank

The DHW tank is modeled as a perfectly stirred control volume with a single state, the water temperature T_{tank} . The energy balance accounts for heat added by the HP and possible auxiliaries (e.g., electric element or solar coil), thermal losses to the surroundings, and the enthalpy removed by draw-offs, as described below:

$$\rho c_p V_{tank} \frac{dT_{tank}}{dt} = Q_{tank}^{hp} + Q_{tank}^{aux} - Q_{dhw}^{dem} - \frac{(T_{tank} - T_{surr})}{R_{tank}} \quad (3)$$

$$Q_{dhw}^{dem} = \frac{\rho c_p V_{dem} (T_{dem} - T_{cold})}{h} \quad (4)$$

where ρ [kg/m³] and c_p [J/(kg K)] denote water density and specific heat, V_{tank} [m³] the tank volume, R_{tank} [K/W] the thermal resistance, T_{surr} [°C] the ambient temperature, V_{dem} [m³] the drawn hot water volume, T_{dem} [°C], the outlet temperature, and T_{cold} [°C] the inlet temperature.

2.3 Equivalent electrical time-series profile

The dynamic coefficient of performance, $COP(\Delta T)$, maps thermal demand to electrical power. The COP mainly depends on the temperature difference ($\Delta T_{hp} = T_{hp}^{supply} - T_{hp}^{source}$) between the source and emitter. This dependence is modeled using the quadratic regression model Fischer et al. (2017), shown in (5), where a_{hp} are HP-specific tunable parameters.

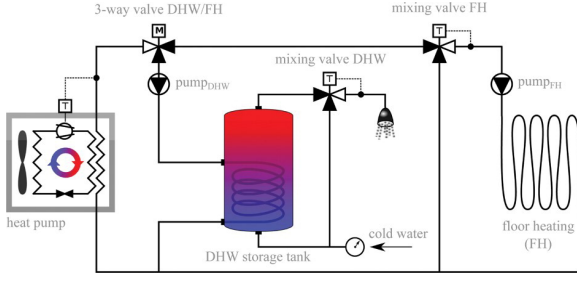


Fig. 1. Hydraulic scheme of the space heating and DHW system De Coninck et al. (2014).

$$COP_{hp} = a_{hp}^0 - a_{hp}^1 \Delta T_{hp} - a_{hp}^2 (\Delta T_{hp})^2 \quad (5)$$

The overall energy consumption is also influenced by supplementary components that run alongside the HP, such as fans and circulation pumps, which consume electricity and increase total demand. Consequently, a constant correction factor (γ_{corr}) is integrated into the COP calculations. The maximum HP capacity depends on the source and supply temperatures, as described in equation (6). Operationally, a three-way selector valve routes the HP's condenser either to the DHW storage tank or the space-heating circuit, so only one branch is served at a time, as depicted in Fig. 1. The mapping equation (7) expresses the electrical power.

$$\bar{Q}_{hp} = Q_{hp}(T_h^{source}, T_h^{supply}) \quad (6)$$

$$P_{hp} = \frac{\min\{Q_{hp}, \bar{Q}_{hp}\}}{COP_{hp}(T_{hp}^{supply}, T_{hp}^{source})\gamma_{corr}} \quad (7)$$

3. FLEXIBILITY QUANTIFICATION IN HP OPERATION

Heat pump flexibility is intrinsically state-dependent. Available upward and downward headroom is bounded by comfort limits, and DHW/tank limits, not only by the unit's capacity. Consequently, system operation must explicitly manage the temperature trajectory as a control variable when activating flexibility, both for load increases and curtailments, so that user comfort preferences are respected. Therefore, the building and the tank are considered as a virtual battery, and the maximum available "charging" and "discharging" energy for SH and DHW is described with the following equations for each time step, where all variables are known from the baseline operation:

$$Q_{sh}^{max} = \frac{C_{in}}{f_h} \frac{dT_{in}^{max}}{dt} - \frac{T_e - T_{in}}{f_h R_{in,e}} - \frac{T_a - T_{in}}{f_h \cdot R_{in,a}} - \frac{A_{in} Q_{sol}}{f_h} \quad (8)$$

$$Q_{tank}^{max} = \rho c_p V_{tank} \frac{dT_{tank}^{max}}{dt} + Q_{dhw}^{dem} + \frac{(T_{tank} - T_{surr})}{R_{tank}} \quad (9)$$

$$Q_{sh}^{min} = \frac{C_{in}}{f_h} \frac{dT_{in}^{min}}{dt} - \frac{T_e - T_{in}}{f_h R_{in,e}} - \frac{T_a - T_{in}}{f_h R_{in,a}} - \frac{A_{in} Q_{sol}}{f_h} \quad (10)$$

$$Q_{tank}^{min} = \rho c_p V_{tank} \frac{dT_{tank}^{min}}{dt} + Q_{dhw}^{dem} + \frac{(T_{tank} - T_{surr})}{R_{tank}} \quad (11)$$

3.1 Upward Flexibility

Upward flexibility denotes the admissible increase in HP electrical power over a short horizon without violating (i) the temperature-dependent capacity of the unit, (ii) indoor comfort limits, or (iii) tank safety/comfort limits. Provided

that the HP is not saturated, headroom exists; however, the effective margin is capped by the most restrictive of these three bounds. During the SH upward flexibility activation, it must be ensured that no DHW requirements exist and vice versa for DHW upward flexibility (12). With mutually exclusive routing via the three-way valve, the thermal headroom is computed by rearranging the building and tank balances, and the corresponding electrical margin is obtained through the COP mapping as described in the following equations for each time-step:

$$0 \leq Q_{sh,t}^{\uparrow} \leq \begin{cases} Q_{sh,t}^{ref} & \text{if } Q_{tank,t}^{min} > 0 \\ Q_{sh,t}^{max} & \text{if } Q_{sh,t}^{min} = 0 \end{cases} \quad (12)$$

$$0 \leq Q_{tank,t}^{\uparrow} \leq \begin{cases} Q_{sh,t}^{ref} & \text{if } Q_{sh,t}^{min} > 0 \\ Q_{sh,t}^{max} & \text{if } Q_{sh,t}^{min} = 0 \end{cases} \quad (13)$$

$$\{Q_{sh,t}^{\uparrow}, Q_{tank,t}^{\uparrow}\} \leq \bar{Q}_{hp,t} \quad (14)$$

$$\{P_{sh,t}^{\uparrow}, P_{tank,t}^{\uparrow}\} = \frac{\{Q_{sh,t}^{\uparrow}, Q_{tank,t}^{\uparrow}\}}{COP_{hp}(T_{hp,t}^{supply}, T_{hp,t}^{source})\gamma_{corr}} \quad (15)$$

$$P_{hp,t}^{\uparrow} = \max\{P_{sh,t}^{\uparrow}, P_{tank,t}^{\uparrow}\} \quad (16)$$

The SH and DHW flexibility quantification is explicitly investigated, as described in detail later, to avoid the use of binary decision variables, which would make the problem intractable by increasing the computational burden and rendering it non-convex.

3.2 Downward Flexibility

Downward flexibility denotes the admissible decrease in HP electrical power relative to the baseline while maintaining indoor comfort and DHW/tank temperature limits, and respecting non-negativity of delivered heat. Similarly to the upward case, the effective margin is governed by the most restrictive bound implied by the building and tank.

$$Q_{sh,t}^{\downarrow} \geq \begin{cases} Q_{sh,t}^{ref} & \text{if } Q_{tank,t}^{min} > 0 \\ Q_{sh,t}^{min} & \text{if } Q_{sh,t}^{min} = 0 \end{cases} \quad (17)$$

$$Q_{tank,t}^{\downarrow} \geq \begin{cases} Q_{sh,t}^{ref} & \text{if } Q_{sh,t}^{min} > 0 \\ Q_{sh,t}^{min} & \text{if } Q_{sh,t}^{min} = 0 \end{cases} \quad (18)$$

$$\{Q_{sh,t}^{\downarrow}, Q_{tank,t}^{\downarrow}\} \geq \bar{Q}_{hp,t} \quad (19)$$

$$\{P_{sh,t}^{\downarrow}, P_{tank,t}^{\downarrow}\} = \frac{\{Q_{sh,t}^{\downarrow}, Q_{tank,t}^{\downarrow}\}}{COP_{hp}(T_{hp,t}^{supply}, T_{hp,t}^{source})\gamma_{corr}} \quad (20)$$

$$P_{hp,t}^{\downarrow} = \min\{P_{sh,t}^{\downarrow}, P_{tank,t}^{\downarrow}\} \quad (21)$$

4. SECURED OPF QUANTIFICATION FRAMEWORK

The proposed framework separates (i) a *local* baseline optimization at each HP prosumer, run with day-ahead price information to compute a cost-minimizing reference schedule, from (ii) a *feeder-level* OPF that quantifies upward/downward flexibility. This architecture minimizes online communication and preserves privacy, while ensuring that thermal-comfort and DHW constraints are enforced locally. Network constraints (voltage, thermal limits) are enforced in the feeder OPF when activating flexibility. The solving algorithm is provided in Algorithm 1.

Algorithm 1 Cost-effective operation and secure-constrained flexibility quantification

Input: Day-ahead prices c_t , loads and DERs forecasts, thermal models of building and tank, network data.

Initialization: Set thermal comfort bounds, tank temperature limits, HP capacity and COP characteristics.

Stage 1: Local Baseline Scheduling

$$\min_{P_{hp,t}^{ref}, \delta_{hp,t}} \sum_{hp \in \Omega^{hp}} \sum_{t \in \Omega^T} c_t P_{hp,t}^{ref} h$$

subject to building/tank dynamics and comfort constraints (20)-(25).

Obtain reference schedule $P_{hp,t}^{ref}, \delta_{hp,t}$.

Stage 2: Flexibility Quantification

for each scenario $s \in \{sh, dhw\}$ **do**

Solve feeder-level OPF for upward envelope:

$$\begin{aligned} \min_{P_{s, hp, t}^\uparrow} \sum_{hp \in \Omega^{hp}} \sum_{t \in \Omega^T} -P_{s, hp, t}^\uparrow \\ \text{s.t. (26) - (30), (31)} \end{aligned}$$

Solve feeder-level OPF for downward envelope:

$$\begin{aligned} \min_{P_{s, hp, t}^\downarrow} \sum_{hp \in \Omega^{hp}} \sum_{t \in \Omega^T} P_{s, hp, t}^\downarrow \\ \text{s.t. (26) - (30), (32)} \end{aligned}$$

end for

Results:

- SH flexibility margins ($P_{hp, sh, t}^\uparrow, P_{hp, sh, t}^\downarrow$),
- DHW flexibility margins ($P_{hp, dhw, t}^\uparrow, P_{hp, dhw, t}^\downarrow$).

Stage 3: Aggregated Flexibility Combination

for each $t \in \Omega^T$ **do**

Compute combined margins: (15), (19)

end for

Output: Baseline HP schedule $P_{hp,t}^{ref}$, and single upward/downward flexibility envelopes.

4.1 Baseline Scheduling Constraints (Stage 1)

The baseline scheduling aims to optimally operate the HPs for SH and DHW provision while satisfying user preferences for indoor and tank temperature, as described in (20). Equations (21) - (22) define the upper and lower limits of HP thermal power. The binary variable $\delta_{hp,t}$ determines the valve position, ensuring that SH and DHW heating do not occur simultaneously. The constraints (24) - (25) define the corresponding temperature bounds. During tapping events, the indicator variable $z_{tank,t}^{tap} = 1$ enforces the comfort condition $T_{tank,t} \geq T_{tank,t}^{dem}$. The following constraints are added for each $hp \in \Omega^{hp}$ and $t \in \Omega^T$:

$$(1), (2), (3), (5), (6), (7) \quad (20)$$

$$Q_{tank,t}^{hp} \geq \delta_{hp,t} \cdot \underline{Q}_{hp,t}, \quad Q_{tank,t}^{hp} \leq \delta_{hp,t} \cdot \bar{Q}_{hp,t} \quad (21)$$

$$Q_{sh,t}^{hp} \geq (1 - \delta_{hp,t}) \cdot \underline{Q}_{hp,t}, \quad Q_{sh,t}^{hp} \leq (1 - \delta_{hp,t}) \cdot \bar{Q}_{hp,t} \quad (22)$$

$$Q_{hp,t} = Q_{tank,t}^{hp} + Q_{sh,t}^{hp} \quad (23)$$

$$T_{in,t}^{min} \leq T_{in,t} \leq T_{in,t}^{max} \quad (24)$$

$$T_{tank,t}^{cold} + z_{tank,t}^{tap} (T_{tank,t}^{dem} - T_{tank,t}^{cold}) \leq T_{tank,t} \leq \bar{T}_{DHW} \quad (25)$$

Thus, the resulting baseline problem is a mixed-integer program that maintains computational efficiency while enabling fast convergence to the optimal solution.

4.2 Power Balance Constraints

Depending on the reference scenario and the maximum upward or downward flexibility, four power flow balances must be satisfied at each time period to guarantee secure operation. The first two ensure safe activation of upward and downward flexibility for SH, while the latter two ensure safe activation of DHW flexibility. Consequently, the following constraints are solved twice, once for the SH case and once for the DHW case. The nodal injections are first computed (26). Then, the network feasibility is enforced using the SOCP branch-flow constraints (27) Panagi et al. (2026b). The following constraints are added for each $s \in \Omega^S$, $k \in \Omega^B$, $t \in \Omega^T$, $flex \in \{\uparrow, \downarrow\}$, and $(k, m) \in \Omega^L$:

$$\begin{aligned} P_{s,k,t}^{inj} &= \sum_{der \in \Omega_k^{der}} \hat{P}_{der,t} - \sum_{hp \in \Omega_k^{hp}} P_{s, hp, t}^{flex} - \sum_{d \in \Omega_k^d} \hat{P}_{d,t}, \\ Q_{s,k,t}^{inj} &= Q_{k,t}^{der} - \sum_{hp \in \Omega_k^{hp}} P_{s, hp, t}^{flex} \tan(\phi_{hp}) - \sum_{d \in \Omega_k^d} \hat{P}_{d,t} \tan(\phi_d) \end{aligned} \quad (26)$$

$$\mathcal{F}_{SOCP} \left(P_{s,k,t}^{inj}, Q_{s,k,t}^{inj}, v_{s,k,t}, \ell_{s,km,t}, P_{s,km,t}, Q_{s,km,t} \right) \leq 0, \quad (27)$$

The branch currents, transformer loading, voltage operational limits, and slack bus assumptions for each $s \in \Omega^S$ and $t \in \Omega^T$ take on the following form:

$$\underline{v}_k \leq v_{s,k,t} \leq \bar{v}_k \quad \forall k \in \Omega^B, \quad v_{s,slack,t} = 1 \quad (28)$$

$$\ell_{s,km,t} \leq \bar{I}_{km}^2, \quad \forall km \in \Omega^L \quad (29)$$

$$\sqrt{(P_{s,t}^{tra})^2 + (Q_{s,t}^{tra})^2} \leq \bar{S}^{tra} \quad (30)$$

4.3 HP Flexibility Constraints

Flexibility constraints are used to ensure operation within the technical and user preferences limits during flexibility activations (see Sections 3.1 and 3.2). The following constraints are added for each $s \in \Omega^S$, $hp \in \Omega^{hp}$ and $t \in \Omega^T$.

$$(12), (13), (14) \quad \text{for upward flexibility} \quad (31)$$

$$(16), (17), (18) \quad \text{for downward flexibility} \quad (32)$$

5. CASE STUDY

The input time series data are illustrated in Fig. 2. The calibrated building model, HP parameters, and regression coefficients are summarized in Table 1, while building assumptions and DHW tank parameters are given in Table 2. An air-source heat pump is assumed for all buildings, justified by Cyprus's mild Mediterranean climate under such conditions Gaur et al. (2021).

5.1 Household Basic Load and PV Generation Profiles

We use real electricity consumption data from 68 residential customers in Cyprus, recorded every 30 minutes over an entire year without any heating system. January is chosen for analysis due to its higher heating demand. Statistical data are used to randomly allocate rooftop PV capacity within each residential building according to the probability mass function provided by the Cyprus DSO. Given the geographic proximity of the households, we assume the PV generation profile is the same, as shown in Fig. 2, but scaled by the installed PV capacity.

Table 1. Model coefficients used for the different heating systems and building

Eq.	Model	T_{supply}	a_0	a_1	a_2
(5)	COP	35°C	3.87	0.098	0.0011
(5)	COP	45°C	3.08	0.079	0.0009
(5)	COP	55°C	2.30	0.056	0.0008
(6)	Capacity	35°C	12.34	0.34	0.0022
(6)	Capacity	45°C	11.81	0.32	0.0020
(6)	Capacity	55°C	9.54	0.27	0.0007
$R_{in,e}$	$R_{in,a}$	$R_{e,a}$	C_{in}	C_e	A_{in}
0.00038	0.04996	0.00307	$4.8 \cdot 10^7$	$8.2 \cdot 10^8$	6.6
A_e	f_h	R_{tank}	V_{tank}	ρ	c_p
7.3	0.87	0.68	0.25	1000	4180

Table 2. Building assumptions and tank parameters

Item	Specification
Heating systems	60% radiators (55°C), 20% fan-coil (45°C), 20% underfloor (35°C)
Weather data	January ambient temperature and solar irradiation (Open-Meteo)
Occupancy	Arrival/departure from mobility survey (probabilistic patterns)
Tank parameters	$R = 0.68 \text{ K/W}$, $V = 250 \text{ L}$, $C_w = 1.05 \text{ MJ/K}$ Reynders et al. (2018)
DHW demand	Stochastic profile sampled every 30 min Ahmed et al. (2016)
DHW Temperature limits	$T_{max} = 65^\circ\text{C}$ (hygiene), $T_{dem} = 55^\circ\text{C}$, $T_{cold} = 10^\circ\text{C}$
HP data	COP and capacity from manufacturer data Toshiba Carrier (2021)

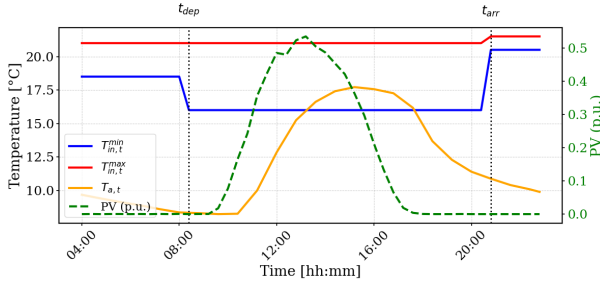


Fig. 2. Input data: indoor comfort limits, ambient temperature, and PV generation over a 24-hour period.

5.2 Distribution Network Model and Price Signals

The proposed methodology is validated on a realistic Cyprus LV DN derived from the publicly available DSO dataset Panagi (2025). The network consists of an 11/0.4 kV, 315 kVA substation transformer and three radial feeders totaling 61 nodes and approximately 30 residential customers per feeder. The dynamic Time-of-Use tariff applied in this study was adopted from ENTSO-E Transparency Platform (2025) for Greece, as Cyprus currently does not operate an electricity market. This dataset provides a realistic representation of dynamic residential tariffs and used as the basis to optimize the reference scenario.

6. RESULTS AND DISCUSSIONS

The optimization problem was solved using Gurobi on an M4 Pro Max chip with a 12-core CPU and 36 GB unified memory. Average computation times were 1.8 min for day-

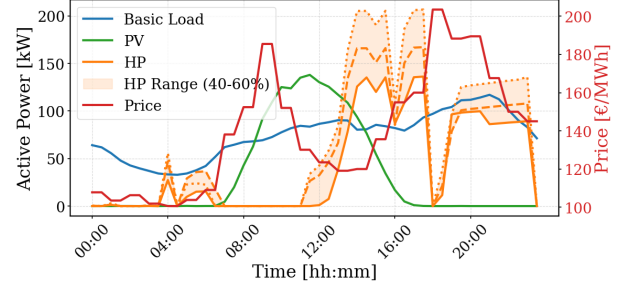


Fig. 3. Reference scenario of HP operation under price-based control, and the interaction with basic load, PV generation, and electricity price throughout the day

ahead scheduling and 5.3 s and 8.4 s for SH and DHW flexibility quantification, respectively.

6.1 Reference Scenario

The aggregate operational power of the HPs, together with the average indoor and tank temperatures, is shown in Fig. 3. Due to price-based operation, significant power peaks occur during low-price hours, while operation is reduced, or even completely curtailed, during high-price periods. Pre-heating and thermal storage are exploited to shift demand and avoid peak prices, while maintaining indoor comfort and DHW availability. At higher penetration levels, peak power increases significantly, although part of this impact is mitigated by coincident PV generation.

6.2 Upward Flexibility Quantification

In Fig. 4, the aggregate time-series active power flexibility for 40%-60% HP penetration is presented. The secured upward flexibility represents the power that can be increased from normal operation without violating user preferences or network constraints. In contrast, the unsecured flexibility reflects the theoretical maximum, reaching the nominal HP power. The upward flexibility power remains high across most periods, but is reduced during peak operation hours due to limited headroom. At all HP penetration levels, the flexibility is restricted during midnight and around noon due to user-preferred temperature limits. Moreover, it is important to note that no network limits are reached for 40% and 50% HP penetration; therefore, no violations occur. However, at 60% HP penetration, a third restriction period emerges during nighttime due to network limits, with the minimum voltage reaching 0.95 p.u. (Fig. 5).

6.3 Downward Flexibility Quantification

The downward flexibility is depicted in Fig. 6 for 40%-60% HP penetration. The unsecured scenario represents the theoretical maximum flexibility, where a deactivation signal switches off HP. Three constrained flexibility periods arise from user-preference limits. The first period occurs after 4:00 while the indoor temperature slightly falls and approaches the limit of 18.5 °C. During the afternoon, another secured, reduced-flexibility period was observed due to the tight indoor constraints during the arrival period. Finally, the most restricted interval occurs at night due to occupancy, although limited flexibility headroom still exists. Network violations are unlikely, as HP deactivation

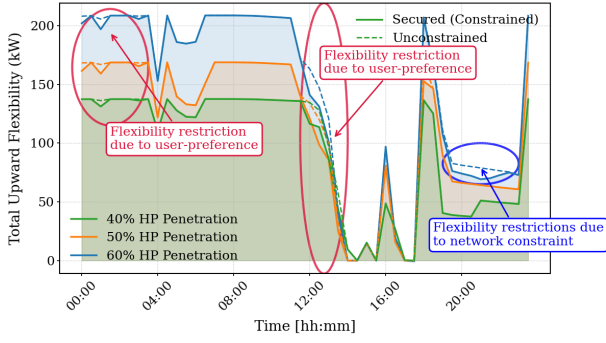


Fig. 4. Total upward flexibility from HPs under different penetration levels (40%, 50%, 60%), comparing secured (constrained) and unconstrained operation.

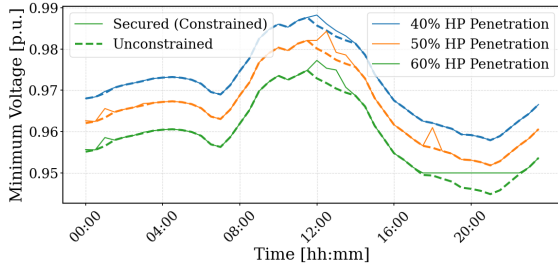


Fig. 5. Minimum voltage profiles under different HP penetration levels (40%, 50%, 60%), comparing secured (constrained) and unconstrained operation.

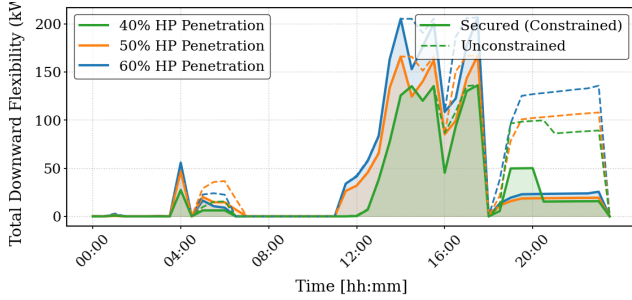


Fig. 6. Total downward flexibility from HPs under different penetration levels (40%, 50%, 60%), comparing secured (constrained) and unconstrained operation.

increases voltage levels and reduces loading. Only extreme reverse power flow conditions may lead to overvoltage.

7. CONCLUSION

This paper presents a constrained flexibility quantification framework that accounts for user preferences and network limits. The results demonstrate that, for upward flexibility, restrictions may arise from both user preferences and network constraints, underscoring the need to consider both. However, at low HP penetration levels, depending on network characteristics, network constraints may not be reached. Therefore, it is crucial to perform a congestion analysis under normal operating conditions first and network constraints may be omitted. Regarding downward flexibility, network constraints are even less likely to be reached, as full HP deactivation generally increases voltage levels and reduces load demand. Future work will extend the framework to real-time operation, integrating model

predictive control to address uncertainties and exploring coordination with other flexible resources such as EVs.

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